

Exercise Sheet 5 - Differential Equations

Deadline for submission: Wednesday, 5 November 2025 at 8:00.

Discussed in exercise session: Friday, 7 November 2025 at 10:15.

chapter1-6

Exercise 5.1 - Second Order Differential Equation

Consider the following **ordinary differential equation (ODE)**:

$$y''(x) - y'(x) - 2y(x) = -2x^2 - 3$$

with the initial conditions:

- a) $y(0) = 0, \quad y'(0) = 0$
- b) $y(0) = 0, \quad y'(0) = 2$
- c) $y(0) = 0, \quad y'(0) = -1.$

1. Compute the homogeneous solution of the differential equation by rewriting it into a system of a first order ODE.
2. Determine its particular solution.
3. Specify the general solution of the ODE for the given initial conditions and visualize it in a figure.

Exercise 5.2 - SEIR: Modeling Epidemics - Part I

In `SEIRS_empty.ipynb` you are provided the implementation of the SEIR model shown in the lecture. In this exercise we will further explore the behavior and small variations of the model. For this, complete the Code provided in the provided notebook.

SEIR: R0 and Vaccinations

1. Set the starting conditions s_0, e_0, i_0, r_0 for our model (remember, that they are the normalized values, i.e. the small letter variables in the slides).
2. Evaluate and plot the model for differing starting times of the lockdown and varying R0s (for both the lockdown-R0 and the unregulated/No-Lockdown R0).
3. Assuming no lockdown interferes and the epidemic naturally proceeds, what effect does the unregulated (No-Lockdown) R0 have on the total number of infected?

4. Expand the model with the assumption that a proportion of the population is vaccinated before the outbreak (and assume the vaccination grants 100% immunity). Hint: You do not have to change the differential equations.
5. What proportion has to be vaccinated to achieve herd immunity and avoid an outbreak? Does it depend on R_0 and if so, can you empirically find a relation?

Exercise 5.3 - SEIR: Modeling Epidemics - Part II

a) SEIR: Multiple Lockdowns

Often, stronger measures for decreasing the target R_0 (lockdowns) are relaxed after some time and have to be reinstated again as soon as the number of cases increases. However, the reintroduction of the measures might be slower, i.e. the transition to the target R_0 can be slower.

1. Change the function `f_seir_two_lds` such that our model has two lockdowns at different times, where the second lockdown has a slower rate of reaching the target (Lockdown) R_0 . You will only need a small modification of the previous `f_seir_ld`.
2. Can you recognize a well-known phenomenon in the number of infected?

b) SI(S): Recovery and Immunity

Consider a disease that does not cause death, but from which infected can not recover. A simple model for this would be given by the following pair of differential equations (the SI Model):

$$\begin{aligned}\frac{ds}{dt} &= -\beta si \\ \frac{di}{dt} &= \beta si\end{aligned}$$

where s and i are again the normalized values of susceptible / infected people. β is a constant.

1. Expand the model in such a way that it allows for recovery that is proportional to the number of infected (with recovery rate α). However, different to the SIR model, one does not obtain any kind of immunity after recovering.
2. Implement and evaluate the model for different parameters in a similar way as the previously provided SEIR model (using `scipy odeint`)